

Visualization 101 2026

Data Visualization In R

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1 Visualization 101 2026

This book is the working text for a five-day course on data visualization in R. It is designed to be read, run, edited, rendered, and reused. The goal is not only to make attractive graphs, but to make visual choices that clarify an argument, reveal a pattern, and leave a reproducible record of how the result was made.

Visualization is a sequence of choices. A plot emphasizes some comparisons and pushes others into the background. It chooses what counts as the main unit, what deserves color or space, what should be labeled directly, what can remain implicit, and what should be excluded because it distracts from the point. The course therefore treats visualization as part of analysis rather than as decoration added at the end.

R is the statistical language used throughout the course. RStudio Desktop and Posit Cloud are the interfaces used to write and run code. The tidyverse is the central programming framework: it supplies a readable grammar for importing, reshaping, summarizing, and plotting data. `ggplot2`, the tidyverse visualization package, is the main plotting system.

Quarto is the document system used for the course files. A Quarto document can contain prose, code, output, figures, tables, citations, and links in one place. The same source document can produce HTML, PDF, slides, dashboards, and reports. When the data or code changes, the outputs can be regenerated from the source instead of manually updated across Word documents, spreadsheets, and slide decks.

Literate programming also supports replication. Code and interpretation live together, which makes it easier to return to a project after a long break, share work with collaborators, and identify exactly how a result was produced. Writing the explanation forces decisions to be made explicit while they are still fresh.

1.1 Schedule

Day	Main Focus	Chapters
1	Workflow, Quarto, data import, and an introduction to <code>ggplot2</code>	2–3, start 4
2	Core <code>ggplot2</code> , separating and comparing data, distributions	finish 4, 5–6

Day	Main Focus	Chapters
3	Lines, bars, annotation, and from question to final figure	7–8
4	Interactivity, dashboards, and tables	9–10
5	Model visualization, mapping, and agentic AI	11–13

Each day has five instructional sessions. The written chapters contain more material than the minimum live path so that faster pacing, review, and post-course study are supported.

1.2 Setup

The Posit Cloud project should already include the packages needed for the main course. The file `install_packages.R` automates installation if the materials are moved to a local computer. The file `install_extra_packages.R` installs optional packages used only in extra sections.

Run `check_setup.R` to verify that the main packages and data files are available.

1.3 Files

The main book chapters are the numbered `.qmd` files in this project. Data files are under `Data/`. Standalone examples are under `Demos/`. Short exercises and solutions are under `Exercises/`.

The course examples use files included with the project, such as `Data/gapminder/gapminder.csv`.

2 Start Here: Workflow, Quarto, And RStudio

2.1 Learning Goals

By the end of this chapter, the main goals are:

- open and navigate the course project
- understand the relationship among R, RStudio, Posit Cloud, and Quarto
- run code chunks inside a `.qmd` file
- render a Quarto document to HTML
- use source mode, visual mode, and the document outline intentionally

2.2 The Course Workflow

This course uses R for computation, RStudio for the working interface, and Quarto for the documents that combine prose, code, and output. The goal is not only to make individual plots. The goal is to build a reproducible workflow where the text explains the question, the code performs the work, and the output records what happened.

During class, most work will happen inside `.qmd` files like this one. A Quarto file is a plain-text document that can contain headings, paragraphs, lists, links, code chunks, tables, figures, and notes about interpretation. When Quarto renders the file, it runs the code chunks and combines the results with the surrounding text.

This is different from working only in the console. The console is useful for quick tests, but it does not automatically preserve the reasoning around the command. A Quarto document keeps the analysis in order: what question was being asked, what data were used, what code was run, what output appeared, and what conclusion followed.

2.3 Plain R Scripts

The simplest kind of R source file is a plain R script. A plain R script uses the `.R` extension and contains only R code. It is useful for setup tasks, repeated checks, helper functions, and other code that does not need surrounding prose or rendered output.

This project includes a few plain R scripts:

```
install_packages.R  
install_extra_packages.R  
check_setup.R
```

The file `check_setup.R` is a good example. It checks whether the main packages and course data files are available. Opening that file and clicking **Source** runs the whole script. The result is a quick setup check, not a rendered document.

Use `.R` scripts when the goal is to run code directly. Use `.qmd` files when the goal is to combine explanation, code, and output.

2.4 The Console

The console is the fastest place to test a command, inspect an object, or rerun something without rendering a document.

Two console habits are especially helpful:

- the up arrow cycles through recent commands
- `Ctrl + R` searches command history in many RStudio setups

The console is not a replacement for a source file. If a command becomes part of the analysis, move it into a `.qmd` or `.R` file so it is preserved.

2.5 R, RStudio, Posit Cloud, And Quarto

These tools have different jobs:

- R is the programming language that reads data, transforms it, estimates models, and makes graphics.
- RStudio is the interface where you edit files, run code, inspect objects, view plots, and render documents.
- Posit Cloud is a browser-based version of the RStudio workflow that avoids many local installation problems.
- Quarto is the publishing system that turns `.qmd` source files into HTML, PDF, Word, slides, dashboards, and books.

R Markdown was the predecessor used in earlier versions of this course. Quarto keeps the same basic idea, but it is now the preferred format for this version of the workshop. If you have used `.Rmd` files before, the transition should feel familiar: a `.qmd` file still mixes text and code. The main difference is that Quarto uses a more general and consistent document system.

2.6 Project Files And Portable Paths

The examples in this course assume that the project has been opened in RStudio or Posit Cloud. Files are organized inside the project in folders such as `Data/`, `Demos/`, and `Exercises/`.

The data examples use file names like these:

```
Data/gapminder/gapminder.csv
Data/penguins/penguins.csv
Demos/presentation-demo.qmd
Exercises/04-scatterplot-solution.R
```

Avoid file names that depend on a personal Desktop, Downloads folder, or another location on one person's computer.

Keeping files inside the project matters because the course should work in Posit Cloud, on another computer, or after being archived for later use.

2.7 The RStudio Layout

RStudio usually has four main panes:

- the source pane, where `.qmd` and `.R` files are edited
- the console, where commands can be typed and run directly
- the environment and history pane, where current objects and previous commands are listed
- the files, plots, packages, help, and viewer pane

The exact pane arrangement can be changed under Global Options. The important habit is to know which pane is doing which job. The source pane is for durable work. The console is for quick tests. The environment pane shows what objects currently exist. The viewer pane is often where rendered HTML output appears.

2.8 Source Mode And Visual Mode

RStudio can edit Quarto documents in source mode or visual mode.

Source mode shows the underlying Markdown and code exactly as written. It is the most transparent mode for learning because it makes headings, chunks, chunk options, links, and lists visible.

Visual mode is a more word-processor-like interface. It can be helpful for editing prose, tables, footnotes, and links. It is still editing the same `.qmd` file, but it hides some of the raw syntax while you write.

Both modes are useful. During this course, source mode is usually the clearest mode for understanding how the document works. Visual mode can be helpful later when revising narrative text.

2.9 Markdown Basics

Quarto uses Markdown for ordinary text formatting. A few patterns cover most needs:

```
# First-level heading

## Second-level heading

Regular paragraph text.

- a bullet point
- another bullet point

**bold text**

*italic text*

[Quarto website](https://quarto.org)
```

The number of `#` symbols controls the heading level. Headings also create the document outline, which makes long lessons easier to navigate.

2.10 Code Chunks

An R code chunk starts with three backticks and `{r}`. A useful chunk also has a short name after `r`, then ends with three backticks.

In the source file, the opening line would be ````{r ch02-code-chunks-1}` and the closing line would be `````. The R code between those lines could be:

```
1 + 1
```

The code inside the chunk can be run interactively, and it can also be run during rendering. In RStudio, you can run a chunk with the green play button or with keyboard shortcuts. You can insert a new chunk from the menu or by typing the chunk delimiters yourself.

If that chunk runs, the result would be:

```
[1] 2
```

Chunks can contain more than one line. For example:

```
x <- 10
y <- 25
x + y
```

The assignment operator `<-` stores a value in an object. In the example above, `x` stores 10 and `y` stores 25. After those objects exist, R can use them in later commands.

2.11 Chunk Names And Navigation

Most real code chunks in these course files have a name. The name appears in the opening line of the chunk. For example, this opening line names a chunk `ch02-first-calculation`:

```
# opening line in the source file:
# ``{r ch02-first-calculation}
```

Chunk names make long documents easier to navigate. In RStudio, the document outline can show both section headings and code chunks, so a descriptive chunk name makes it easier to jump to a specific part of the file. Chunk names also make error messages easier to interpret because Quarto can report which chunk failed.

Chunk names should be short, unique, and descriptive. A name such as `ch04-first-scatterplot` is more useful than a name such as `chunk1`.

2.12 Chunk Options

Chunk options control how code behaves during rendering. In Quarto, options are often written as special comment lines at the top of the chunk.

This chunk runs but hides the code:

```
[1] "2026-05-15"
```

This chunk displays code without running it:

```
Sys.Date()
```

`eval: false` is important for examples that require credentials, internet access, private files, or intentional manual action. The code remains visible as a template, but it does not run automatically.

2.13 Rendering

Rendering turns a source `.qmd` file into an output document. For this course, HTML is the primary output because it is easy to view in a browser and works well for code, figures, tables, and links.

When you render a document, Quarto starts from the top, runs the chunks in order, and inserts the output where it belongs. If a chunk fails, rendering stops. That is useful because it exposes broken code early.

The render button in RStudio is the easiest way to start. Open the file and click **Render** to create the output document. PDF output is also useful for papers, handouts, and print-ready reports, but it depends on a working TeX installation. HTML will be the most reliable output during the workshop.

2.14 A First Reproducible Example

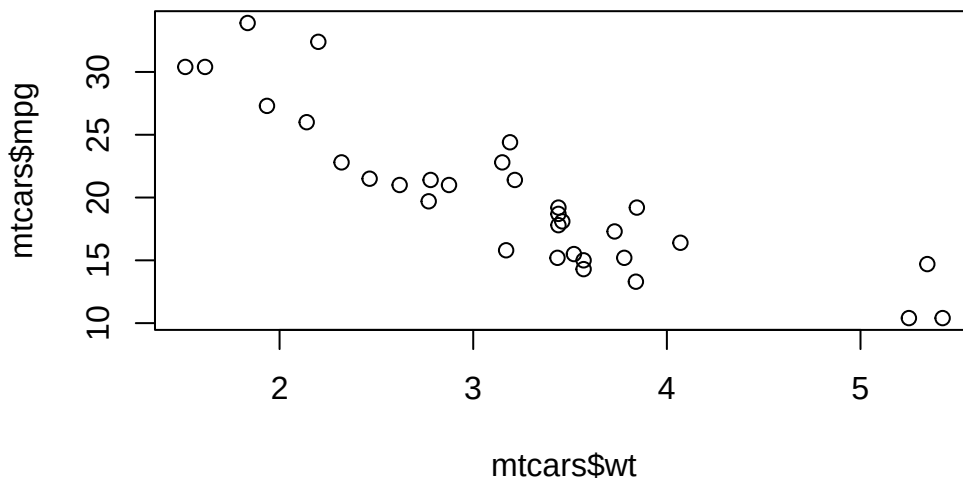
The built-in `mtcars` dataset is available in base R, so it is useful for a first test that does not depend on course data files.

```
head(mtcars)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

This command creates a simple base R plot:

```
plot(mtcars$wt, mtcars$mpg)
```



The plot is not polished, but it confirms that R is running, code chunks are executing, and graphical output appears in the document.

2.15 Reading Output Carefully

A rendered document is not automatically correct just because it renders. Rendering answers one question: did the code run from top to bottom? You still have to ask whether the data are the right data, whether the variables mean what you think they mean, whether missing values were handled correctly, and whether the plot supports the claim being made.

This course will return to that distinction throughout the week. Reproducibility is necessary, but it is not enough. A reproducible mistake is still a mistake.

2.16 Short Exercise

Create a new code chunk below this paragraph. In that chunk:

1. store the number 100 in an object named `start_value`
2. store the number 25 in an object named `change_value`
3. subtract `change_value` from `start_value`

```
# Write your code here.
```

2.17 Extras: RStudio Settings

A few RStudio settings can make workshop work more comfortable:

- turn off restoring `.RData` into the workspace at startup
- set saving the workspace on exit to “Never”
- turn on rainbow parentheses if available
- use the document outline for long `.qmd` files
- choose a source editor font and size that are comfortable for several hours of reading

These settings do not change the analysis, but they reduce friction. The main principle is that the project files should define the work, not a hidden workspace image from a previous session.

2.18 Extras: Output Formats

Quarto can produce many output formats from similar source files:

- HTML documents and books
- PDF reports
- Word documents
- RevealJS slides
- dashboards
- websites

The same analysis can therefore be adapted for different audiences. A rough exploration may become a class exercise. A refined figure may become part of a report. A report may become a short presentation. Quarto is useful because those outputs can remain connected to the same source code and data workflow.

2.18.1 Presentation Demo

Quarto can render slides. A presentation file uses a different YAML header:

```
---
title: "Presentation Demo"
format:
  revealjs:
    theme: simple
---
```

The course includes a standalone revealjs demo at `Demos/presentation-demo.qmd`. The main difference from a book chapter is that second-level headings become slides. Code chunks work the same way, but slides usually hide more code and show more finished output.

```
## Slide Title

Slide text goes here.
```

2.18.2 Paper Or Report Demo

A paper-style Quarto document emphasizes prose, citations, figures, tables, and cross-references. The course includes `Demos/paper-example.qmd` and `Demos/paper-example.bib`. The `.bib` file stores citation records; the `.qmd` file cites them with keys such as `[@wickham_ggplot2_2016]`.

A minimal paper-style header looks like this:

```
---
title: "Paper Demo"
author: "Name"
bibliography: paper-example.bib
format:
  html: default
  pdf: default
---
```

Paper-style files usually keep more prose visible than slides or dashboards. They also make more use of citations, captions, footnotes, and cross-references.

2.19 What Comes Next

The next chapter begins the data workflow: importing files, inspecting tibbles, reshaping data, and using the core tidyverse verbs that prepare data for visualization.

3 Data Import And The Tidyverse

3.1 Learning Goals

By the end of this chapter, the main goals are:

- read local data included with the project
- recognize the difference between a data frame and a tibble
- inspect a dataset before plotting it
- distinguish wide data from long data
- reshape data with `pivot_longer()` and `pivot_wider()`
- use the core tidyverse verbs for common data preparation tasks
- recognize which import examples should not run automatically because they require credentials or external services

3.2 Load Packages

The **tidyverse** is a collection of packages for importing, transforming, and visualizing data. Loading it makes packages such as **readr**, **dplyr**, **tidyr**, **ggplot2**, **stringr**, and **tibble** available.

```
library(tidyverse)
library(skimr)
```

3.3 Importing Local Data

Many projects begin with a local CSV file. CSV files are plain-text tables where columns are separated by commas. They are common because nearly every spreadsheet, statistical package, and database can export them.

The preferred tidyverse function is `read_csv()` from the **readr** package. Compared with base R's `read.csv()`, it usually prints more clearly, guesses column types more consistently, and returns a tibble.

The course data live under `Data/`. The example below reads the Gapminder CSV file included with the project.

```
gap <- read_csv("Data/gapminder/gapminder.csv", show_col_types = FALSE)
```

```
gap
```

```
# A tibble: 1,704 x 6
```

	country	continent	year	lifeExp	pop	gdpPercap
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Afghanistan	Asia	1952	28.8	8425333	779.
2	Afghanistan	Asia	1957	30.3	9240934	821.
3	Afghanistan	Asia	1962	32.0	10267083	853.
4	Afghanistan	Asia	1967	34.0	11537966	836.
5	Afghanistan	Asia	1972	36.1	13079460	740.
6	Afghanistan	Asia	1977	38.4	14880372	786.
7	Afghanistan	Asia	1982	39.9	12881816	978.
8	Afghanistan	Asia	1987	40.8	13867957	852.
9	Afghanistan	Asia	1992	41.7	16317921	649.
10	Afghanistan	Asia	1997	41.8	22227415	635.

```
# i 1,694 more rows
```

The object name `gap` now points to a table in memory. The assignment operator `<-` reads as “store the result on the right in the object named on the left.” Object names should be short enough to type but descriptive enough to understand later.

3.4 Tibbles And First Inspection

A tibble is the tidyverse version of a data frame. It is still a rectangular table with rows and columns, but it prints in a more controlled way. A tibble tells you the number of rows and columns, shows variable types, and avoids flooding the console with every row of a large dataset.

Start every new dataset with a quick inspection:

```
glimpse(gap)
```

```
Rows: 1,704
```

```
Columns: 6
```

```
$ country    <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", ~
```

```
$ continent <chr> "Asia", "Asia", "Asia", "Asia", "Asia", "Asia", "Asia", "Asi~
$ year      <dbl> 1952, 1957, 1962, 1967, 1972, 1977, 1982, 1987, 1992, 1997, ~
$ lifeExp   <dbl> 28.801, 30.332, 31.997, 34.020, 36.088, 38.438, 39.854, 40.8~
$ pop       <dbl> 8425333, 9240934, 10267083, 11537966, 13079460, 14880372, 12~
$ gdpPercap <dbl> 779.4453, 820.8530, 853.1007, 836.1971, 739.9811, 786.1134, ~
```

```
skim(gap)
```

Table 3.1: Data summary

Name	gap
Number of rows	1704
Number of columns	6
Column type frequency:	
character	2
numeric	4
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
country	0	1	4	24	0	142	0
continent	0	1	4	8	0	5	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1	1979.50	17.27	1952.00	1965.75	1979.50	1993.25	2007.0	
lifeExp	0	1	59.47	12.92	23.60	48.20	60.71	70.85	82.6	
pop	0	1	29601212	33615789	60011.00	2793664.00	23595.50	585221.75	18683096.0	
gdpPer-cap	0	1	7215.33	9857.45	241.17	1202.06	3531.85	9325.46	113523.1	

`glimpse()` comes from `dplyr`, which is loaded as part of the tidyverse. It gives a compact structural view of the data. `skim()` comes from the `skimr` package and gives a broader summary, including missingness and descriptive statistics. These checks are not just housekeeping. They help identify the unit of observation, variable types, unexpected missing values, and suspicious ranges before those problems enter a plot.

3.5 Base R Import And Inspection

Base R can also import data. The syntax is common in older examples and in some online answers:

```
gap_base <- read.csv("Data/gapminder/gapminder.csv")
str(gap_base)
head(gap_base)
```

3.6 Other Local File Types

CSV is only one possible format. Social science projects often use Excel, Stata, SPSS, or SAS files. The syntax below should be treated as a template. These chunks are non-executing unless the named files are present and the relevant packages are installed.

3.6.1 Excel

The function `read_excel()` comes from the `readxl` package.

```
library(readxl)

gap_excel <- read_excel(
  path = "Data/gapminder/gapminder.xlsx",
  sheet = "gapminder"
)

gap_excel
```

3.6.2 Stata

The functions for Stata, SPSS, and SAS files below come from the `haven` package.

```
library(haven)

state_legislatures <- read_dta("Data/state_policy/shor_mccarty_state_data.dta")

state_legislatures
```

3.6.3 SPSS

The example uses a small SPSS file that is installed with the **haven** package.

```
library(haven)

path <- system.file("examples", "iris.sav", package = "haven")
survey_data <- read_sav(path)

survey_data
```

3.6.4 SAS

The example uses a small SAS file that is installed with the **haven** package.

```
library(haven)

path <- system.file("examples", "iris.sas7bdat", package = "haven")
admin_data <- read_sas(path)

admin_data
```

3.7 Wide And Long Data

Data can be organized in more than one shape. A *wide* dataset has one row per unit and a separate column for each measurement (`gdpPercap`, `lifeExp`). A *long* dataset stacks the measurement names into one column and the values into another. Wide data is often easier for people to read; long data is often easier for `ggplot2` and group-wise summaries because the measurement name itself can be mapped to color, facets, or groups.

`pivot_longer()` and `pivot_wider()` from `tidyr` move data between the two shapes.

```
wide_data <- tibble(
  country = c("Afghanistan", "Brazil"),
  year = c(2007, 2007),
  gdpPercap = c(974.6, 9065.8),
  lifeExp = c(43.8, 72.4)
)

long_data <- wide_data |>
```

```

pivot_longer(
  cols = c(gdpPercap, lifeExp),
  names_to = "variable",
  values_to = "value"
)

long_data

```

```

# A tibble: 4 x 4
  country      year variable  value
  <chr>      <dbl> <chr>    <dbl>
1 Afghanistan 2007 gdpPercap 975.
2 Afghanistan 2007 lifeExp  43.8
3 Brazil       2007 gdpPercap 9066.
4 Brazil       2007 lifeExp  72.4

```

```

long_data |>
  pivot_wider(names_from = variable, values_from = value)

```

```

# A tibble: 2 x 4
  country      year gdpPercap lifeExp
  <chr>      <dbl>    <dbl>    <dbl>
1 Afghanistan 2007      975.     43.8
2 Brazil       2007     9066.     72.4

```

The pipe operator `|>` means “take the result so far and send it into the next function.” It lets a sequence of data steps read in the same order as the logic of the analysis.

3.8 Core Tidyverse Verbs

Most data preparation before plotting uses a small set of verbs:

- `filter()` keeps rows
- `select()` keeps columns
- `mutate()` creates or modifies variables
- `group_by()` defines groups
- `summarize()` collapses rows into summaries
- joins combine information from two tables

These verbs come from `dplyr`, which is loaded as part of the tidyverse. They are useful because they correspond to ordinary analytical questions. Which cases belong in this plot? Which variables matter? Do we need a new rate, logged value, or category? What is the summary by group? Does another table contain information that should be merged in?

3.9 `filter()`: Keep Rows

`filter()` keeps observations that satisfy logical conditions. The operator `==` tests exact equality. The operator `%in%` tests whether a value belongs to a set.

```
gap |>
  filter(year == 2007)
```

```
# A tibble: 142 x 6
  country    continent year lifeExp      pop gdpPercap
  <chr>      <chr>    <dbl>  <dbl>    <dbl>    <dbl>
1 Afghanistan Asia      2007   43.8  31889923    975.
2 Albania    Europe    2007   76.4   3600523   5937.
3 Algeria    Africa    2007   72.3  33333216   6223.
4 Angola     Africa    2007   42.7  12420476   4797.
5 Argentina  Americas  2007   75.3  40301927  12779.
6 Australia  Oceania   2007   81.2  20434176  34435.
7 Austria    Europe    2007   79.8   8199783   36126.
8 Bahrain    Asia      2007   75.6   708573    29796.
9 Bangladesh Asia      2007   64.1 150448339   1391.
10 Belgium   Europe    2007   79.4  10392226  33693.
# i 132 more rows
```

```
gap |>
  filter(continent %in% c("Americas", "Europe"))
```

```
# A tibble: 660 x 6
  country    continent year lifeExp      pop gdpPercap
  <chr>      <chr>    <dbl>  <dbl>    <dbl>    <dbl>
1 Albania Europe    1952   55.2  1282697   1601.
2 Albania Europe    1957   59.3  1476505   1942.
3 Albania Europe    1962   64.8  1728137   2313.
4 Albania Europe    1967   66.2  1984060   2760.
5 Albania Europe    1972   67.7  2263554   3313.
6 Albania Europe    1977   68.9  2509048   3533.
```

```

7 Albania Europe      1982    70.4 2780097    3631.
8 Albania Europe      1987    72   3075321    3739.
9 Albania Europe      1992    71.6 3326498    2497.
10 Albania Europe     1997    73.0 3428038    3193.
# i 650 more rows

```

Use `==` for one value. Use `%in%` when several values are acceptable.

3.10 `select()`: Keep Columns

`select()` keeps variables. This is useful when a dataset has many columns but the next step only needs a few.

```

gap |>
  select(country, continent, year, lifeExp, gdpPercap)

```

```

# A tibble: 1,704 x 5
  country      continent year lifeExp gdpPercap
  <chr>        <chr>    <dbl>  <dbl>    <dbl>
1 Afghanistan Asia      1952    28.8     779.
2 Afghanistan Asia      1957    30.3     821.
3 Afghanistan Asia      1962    32.0     853.
4 Afghanistan Asia      1967    34.0     836.
5 Afghanistan Asia      1972    36.1     740.
6 Afghanistan Asia      1977    38.4     786.
7 Afghanistan Asia      1982    39.9     978.
8 Afghanistan Asia      1987    40.8     852.
9 Afghanistan Asia      1992    41.7     649.
10 Afghanistan Asia      1997    41.8     635.
# i 1,694 more rows

```

Rows are observations. Columns are variables. `filter()` works on rows; `select()` works on columns.

3.11 `mutate()`: Create Variables

`mutate()` creates new variables or rewrites existing ones. In the next example, `round()` makes GDP per capita easier to read, and `log10()` creates a transformed variable that will be useful for plotting.

```
gap |>
  mutate(
    gdp_pc_round = round(gdpPercap),
    log_gdp_pc = log10(gdpPercap)
  ) |>
  select(country, gdpPercap, gdp_pc_round, log_gdp_pc)
```

```
# A tibble: 1,704 x 4
  country      gdpPercap gdp_pc_round log_gdp_pc
  <chr>         <dbl>         <dbl>     <dbl>
1 Afghanistan  779.           779       2.89
2 Afghanistan  821.           821       2.91
3 Afghanistan  853.           853       2.93
4 Afghanistan  836.           836       2.92
5 Afghanistan  740.           740       2.87
6 Afghanistan  786.           786       2.90
7 Afghanistan  978.           978       2.99
8 Afghanistan  852.           852       2.93
9 Afghanistan  649.           649       2.81
10 Afghanistan 635.           635       2.80
# i 1,694 more rows
```

`round(gdpPercap)` rounds each value in the `gdpPercap` column. `log10(gdpPercap)` takes the base-10 logarithm. The original variable remains unless you explicitly overwrite it.

3.12 `group_by()` And `summarize()`: Aggregate Data

Many plots need data at a different level than the imported file. For example, the Gapminder data are country-year observations, but we may want a continent-level summary.

```
continent_summary <- gap |>
  group_by(continent) |>
  summarize(
    median_life_exp = median(lifeExp, na.rm = TRUE),
    median_gdp_pc = median(gdpPercap, na.rm = TRUE),
    n_countries = n_distinct(country)
  )

continent_summary
```



```
# A tibble: 5 x 4
  continent median_life_exp median_gdp_pc n_countries
  <chr>          <dbl>          <dbl>          <int>
1 Africa          47.8          1192.           52
2 Americas        67.0          5466.           25
3 Asia            61.8          2647.           33
4 Europe          72.2          12082.          30
5 Oceania         73.7          17983.           2
```

`na.rm = TRUE` tells functions such as `median()` to ignore missing values. Without it, one missing value can make a summary missing.

3.13 Readable Pipelines

Long pipelines are easier to read when each verb gets its own line and arguments are indented consistently.

```
gap |>
  filter(year == 2007) |>
  mutate(gdp_pc_round = round(gdpPercap)) |>
  group_by(continent) |>
  summarize(
    median_life_exp = median(lifeExp, na.rm = TRUE),
    median_gdp_pc = median(gdp_pc_round, na.rm = TRUE)
  ) |>
  arrange(desc(median_life_exp))
```

```
# A tibble: 5 x 3
  continent median_life_exp median_gdp_pc
  <chr>          <dbl>          <dbl>
1 Oceania          80.7          29810
2 Europe           78.6          28054
3 Americas         72.9           8948
4 Asia            72.4           4471
5 Africa           52.9           1452
```

The line breaks are not just style. They make the data workflow visible. Each line is one transformation.

3.14 Joins

Joins combine two tables using one or more key variables. A common pattern is to keep all rows from the main table and add matching columns from a smaller lookup table. That is a `left_join()`, another `dplyr` function.

```
region_notes <- tibble(
  continent = c("Africa", "Americas", "Asia", "Europe", "Oceania"),
  example_note = c(
    "Many countries, large range of outcomes",
    "Includes North, Central, and South America",
    "Large population range",
    "High median life expectancy in many examples",
    "Small number of countries in many teaching datasets"
  )
)

continent_summary |>
  left_join(region_notes, by = "continent")
```

```
# A tibble: 5 x 5
  continent median_life_exp median_gdp_pc n_countries example_note
  <chr>          <dbl>          <dbl>      <int> <chr>
1 Africa          47.8          1192.         52 Many countries, large ran~
2 Americas        67.0          5466.         25 Includes North, Central, ~
3 Asia           61.8          2647.         33 Large population range
4 Europe          72.2         12082.         30 High median life expectan~
5 Oceania         73.7         17983.          2 Small number of countries~
```

Rows without a match in the lookup table would remain, but the new variables would be NA. That behavior is usually safer than silently dropping observations.

3.15 Short Exercise

Use the built-in `mtcars` dataset. Write a pipeline that:

1. converts `mtcars` to a tibble with row names stored as a column named `car`
2. groups by `cyl`
3. calculates average `mpg` and maximum `wt`
4. returns one row per value of `cyl`

```
# Write your code here.
```

3.16 Extras: Google Sheets

Google Sheets can be useful when data are collaboratively maintained. These examples require authentication and should not run automatically in a rendered course book.

```
library(google Sheets4)

sheet_data <- read_sheet(
  ss = "PASTE-GOOGLE-SHEET-ID-OR-URL-HERE",
  sheet = "Sheet1"
)

sheet_data
```

```
library(google Sheets4)

ss <- gs4_create("Visualization 101 2026 Example Sheet")

mtcars |>
  as_tibble(rownames = "car") |>
  sheet_write(ss = ss, sheet = "mtcars")
```

3.17 Census Data With tidycensus

Some data sources are best accessed through specialized packages. These examples are templates, not automatically running chunks.

The `tidycensus` package wraps the U.S. Census Bureau's APIs. The two most common entry points are the American Community Survey (`get_acs()`) and the decennial census (`get_decennial()`). A free Census API key is required. Request one at https://api.census.gov/data/key_signup.html, then register it once on your machine with `census_api_key("YOUR_KEY", install = TRUE)`. After that, the key is available to every R session and the line can be removed from teaching files.

```
library(tidycensus)

#census_api_key("2c5ba48be91062db570c7fc5fa49dbca03306c33", install = TRUE)
```

`load_variables()` lists the variables available in a given dataset and year. Caching the result avoids re-downloading the table.

```
acs_vars <- load_variables(2020, "acs5", cache = TRUE) |>
  distinct(concept, .keep_all = TRUE)

acs_vars
```

If `load_variables()` errors with a JSON parse message like `invalid char in json text. <html>`, the Census Bureau returned an HTML error page instead of the variable table. That usually means the requested year is not yet published for that dataset, the Census server is temporarily down, or the request was rate-limited. Try a slightly older year (for ACS 5-year, two years behind the current year is usually safe), re-run the call after a minute, or check <https://www.census.gov/data/developers/data-sets.html> for the dataset's release schedule.

The `variables` argument of `get_acs()` accepts named variable codes. Naming the entries gives readable column names instead of raw codes like `B19013_001`. `output = "wide"` puts each variable in its own column, which is usually easier to plot from than the default long format.

```
acs_raw <- get_acs(
  geography = "congressional district",
  variables = c(
    median_income = "B19013_001",
    total_pop = "B01003_001"
  ),
  year = 2022,
  survey = "acs5",
  output = "wide"
)

acs_raw
```

The `GEOID` column is a concatenation of geographic identifiers. For congressional districts it combines the two-digit state FIPS code with the two-digit district number. Splitting it with `str_sub()` gives joinable columns.

```
acs <- acs_raw |>
  mutate(
    fips = str_sub(GEOID, 1, 2),
    cd = str_sub(GEOID, 3, 4)
  )
```

```
acs
```

State-level summaries are simpler because the `GE0ID` is just the state FIPS code. The wide format makes it easy to compare one variable against another.

```
acs_state <- get_acs(  
  geography = "state",  
  variables = c(  
    median_income = "B19013_001",  
    median_rent = "B25064_001"  
  ),  
  year = 2022,  
  survey = "acs5",  
  output = "wide"  
)  
  
acs_state |>  
  summarize(  
    n_states = n(),  
    cor_income_rent = cor(median_incomeE, median_rentE, use = "complete.obs")  
  )
```

Note that the wide variable names get an E suffix for estimates and an M suffix for the margin of error. Both are usually returned. For most teaching purposes the E columns are enough; in published work the M columns should be acknowledged.

The decennial census uses `get_decennial()` instead. It returns counts rather than estimates and does not need a `survey` argument.

```
pop_2020 <- get_decennial(  
  geography = "state",  
  variables = c(total_pop = "P1_001N"),  
  year = 2020,  
  sumfile = "p1"  
)  
  
pop_2020
```

3.18 Extras: Qualtrics And Canvas

Survey platforms and learning management systems are also possible sources of data. The main point is that API-based imports should be treated as retrieval steps, not as code that runs every time a teaching document renders.

```
library(qualtrics)

raw_survey <- fetch_survey(
  surveyID = "YOUR-SURVEY-ID",
  label = FALSE,
  convert = FALSE
)
```

```
library(rcanvas)

courses <- get_course_list()
```

3.19 What Comes Next

The next chapter uses these data ideas to build plots. `ggplot2` works best when the data are already in a clear rectangular form and the variables needed for the plot have been inspected, selected, transformed, or summarized.

4 Core ggplot2

4.1 Learning Goals

By the end of this chapter, the main goals are:

- compare base R graphics with `ggplot2`
- understand the grammar of graphics at a practical level
- build plots from `ggplot()`, `aes()`, and geoms
- use the `+` operator to build a visualization pipeline
- distinguish mapped aesthetics from fixed aesthetics
- add scatterplot layers, smoothers, log scales, and labels
- apply built-in themes and color palettes

4.2 Load Packages

`ggplot2` is part of the tidyverse. The `scales` package is useful for readable axis labels. `RColorBrewer` provides named color palettes.

```
library(tidyverse)
library(scales)
library(RColorBrewer)
```

4.3 Data For This Chapter

The main examples use the local Gapminder country-year dataset in `Data/gapminder/gapminder.csv`.

```
gap <- read_csv("Data/gapminder/gapminder.csv", show_col_types = FALSE)

gap_2007 <- gap |>
  filter(year == max(year, na.rm = TRUE))

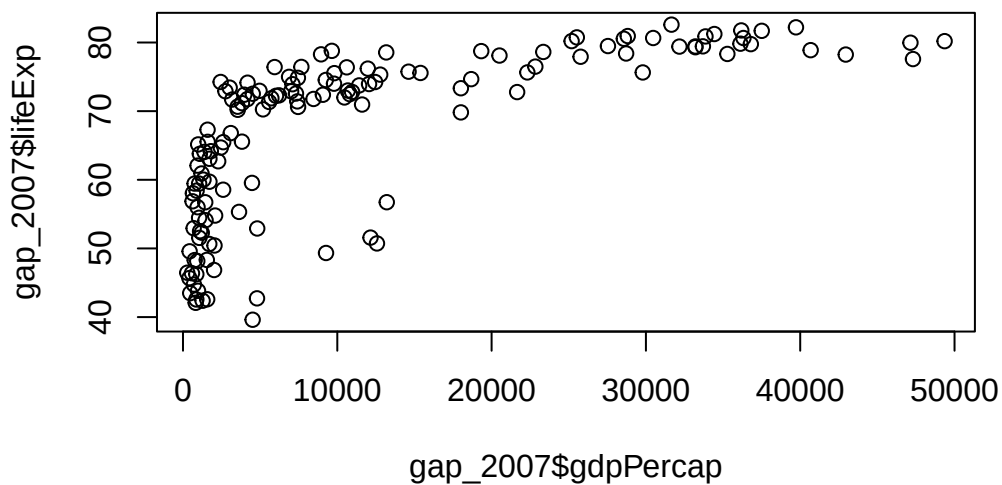
gap_2007
```

```
# A tibble: 142 x 6
  country    continent year lifeExp      pop gdpPercap
  <chr>      <chr>    <dbl>  <dbl>    <dbl>    <dbl>
1 Afghanistan Asia      2007   43.8  31889923    975.
2 Albania    Europe    2007   76.4   3600523   5937.
3 Algeria    Africa    2007   72.3  33333216   6223.
4 Angola     Africa    2007   42.7  12420476   4797.
5 Argentina  Americas  2007   75.3  40301927  12779.
6 Australia  Oceania    2007   81.2  20434176  34435.
7 Austria    Europe    2007   79.8   8199783  36126.
8 Bahrain    Asia      2007   75.6    708573  29796.
9 Bangladesh Asia      2007   64.1 150448339   1391.
10 Belgium   Europe    2007   79.4  10392226  33693.
# i 132 more rows
```

4.4 Comparing Base R Graphics And ggplot2

Base R has plotting functions built in. They are useful for quick checks and appear in many older examples.

```
plot(gap_2007$gdpPercap, gap_2007$lifeExp)
```



That plot works, but the labels are not very informative, the x-axis is hard to read, and customization quickly becomes a matter of remembering many arguments.

A base R histogram shows a similar pattern:

```
hist(gap_2007$lifeExp, breaks = 8)
```



The problem is not that base R cannot make plots. The problem is that complex plots become difficult to revise. `ggplot2` gives us a more structured way to build visualizations.

4.5 The Grammar Of Graphics

A `ggplot2` figure is built in layers. Every plot needs three minimum elements before any layer can sit on top of them:

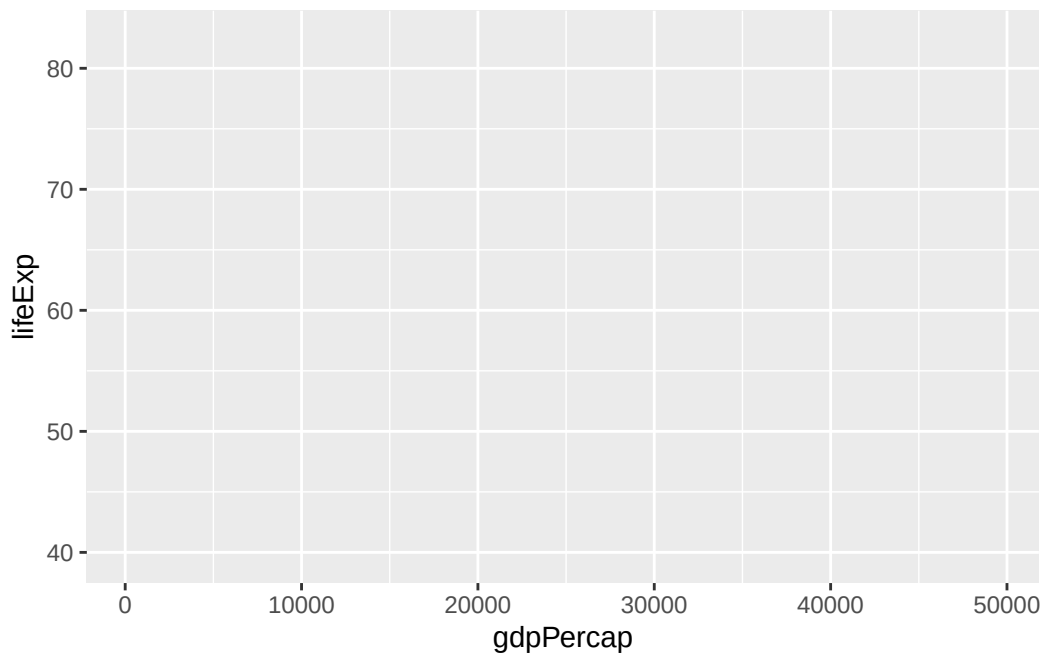
- data: the table being plotted
- mappings: the links between variables and visual features
- geoms: the visual marks, such as points, lines, bars, or histograms

This separation is what makes `ggplot2` powerful. You can change the data, the mapping, or the geom without rewriting the whole plot from scratch.

4.6 The Base Layer: `ggplot()` And `aes()`

The `ggplot()` function starts a plot. The `aes()` function defines aesthetic mappings. Aesthetic mappings connect variables in the data to visual properties such as x-position, y-position, color, size, shape, or transparency.

```
ggplot(  
  data = gap_2007,  
  mapping = aes(x = gdpPercap, y = lifeExp)  
)
```

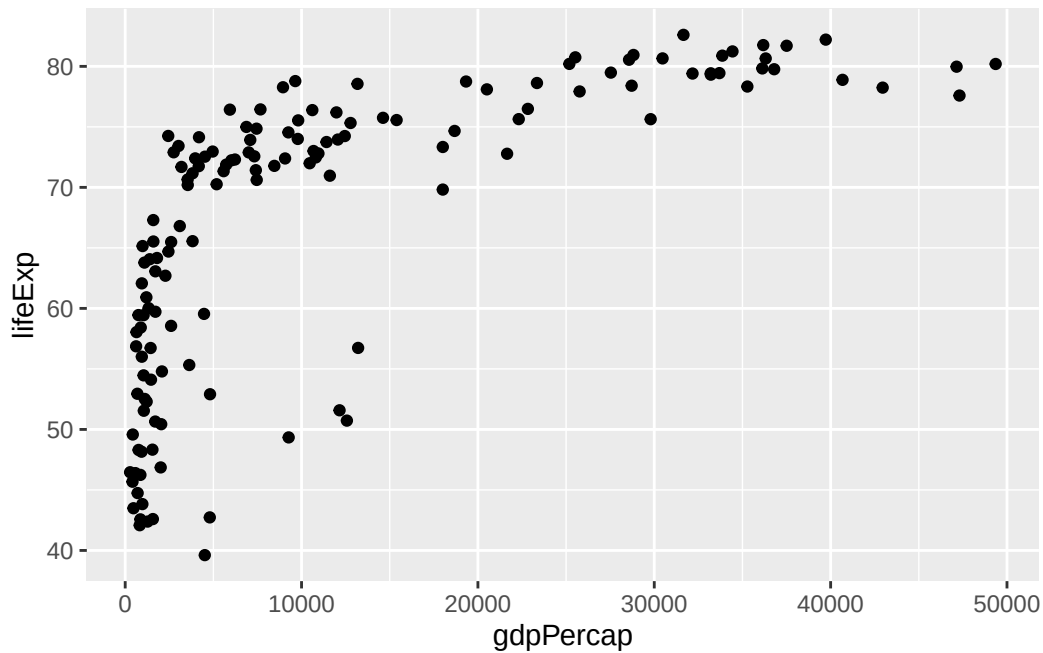


This is the **base layer**. The data and mappings are set, the axes are drawn, but nothing is plotted yet because no geom has been added. Every later layer — points, lines, smoothers, labels, scales — sits on top of this base.

4.7 Add A Geom Layer

`geom_point()` draws points. Because each row in `gap_2007` is a country, each point is a country.

```
ggplot(  
  data = gap_2007,  
  mapping = aes(x = gdpPercap, y = lifeExp)  
) +  
  geom_point()
```

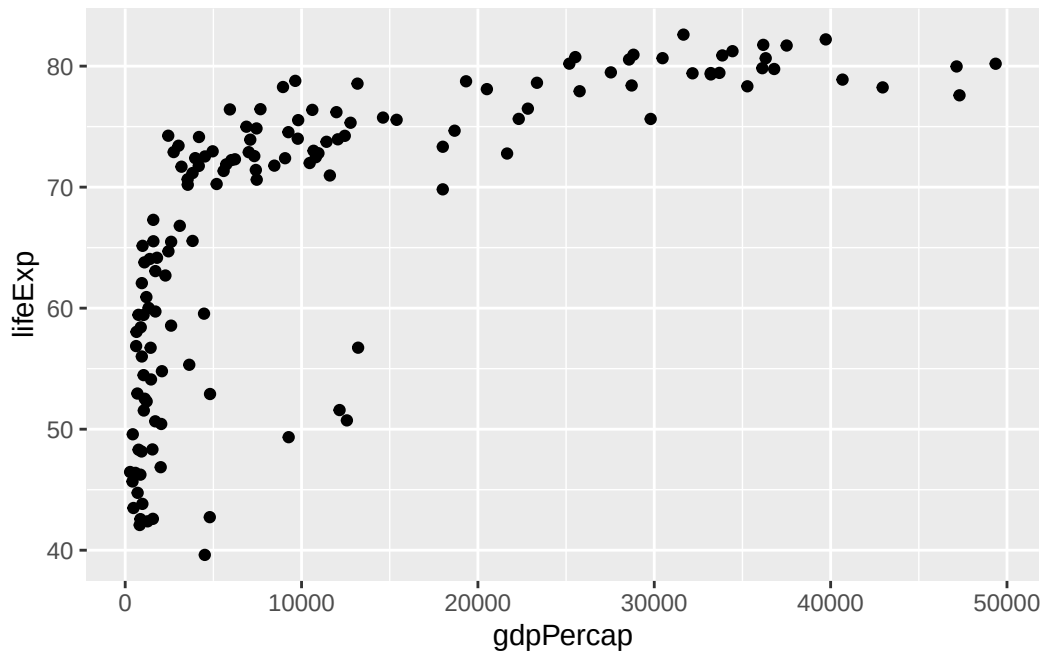


The + operator means “add the next plot component.” In `ggplot2`, line breaks usually put + at the end of the line so R knows the plot is not finished.

4.8 Saving A Plot Object

It is often useful to store the base plot and add layers later.

```
p_gap <- ggplot(  
  data = gap_2007,  
  mapping = aes(x = gdpPercap, y = lifeExp)  
)  
  
p_gap +  
  geom_point()
```



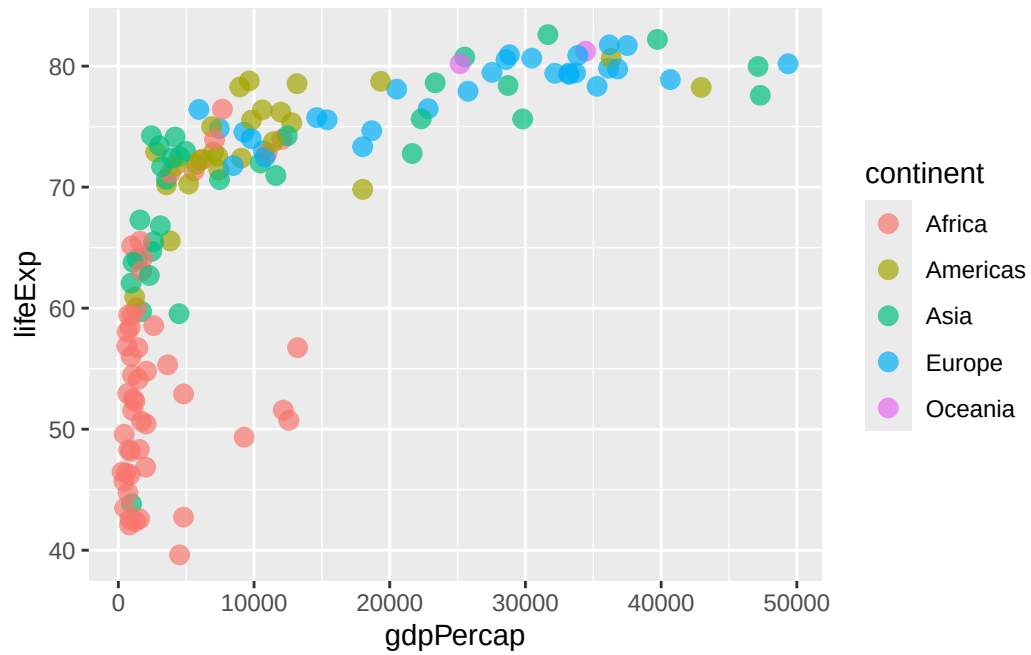
The object `p_gap` stores the data and default mappings. It does not lock the plot into one final form. We can keep adding layers.

4.9 Mapping Versus Setting

One of the most important `ggplot2` distinctions is the difference between mapping and setting.

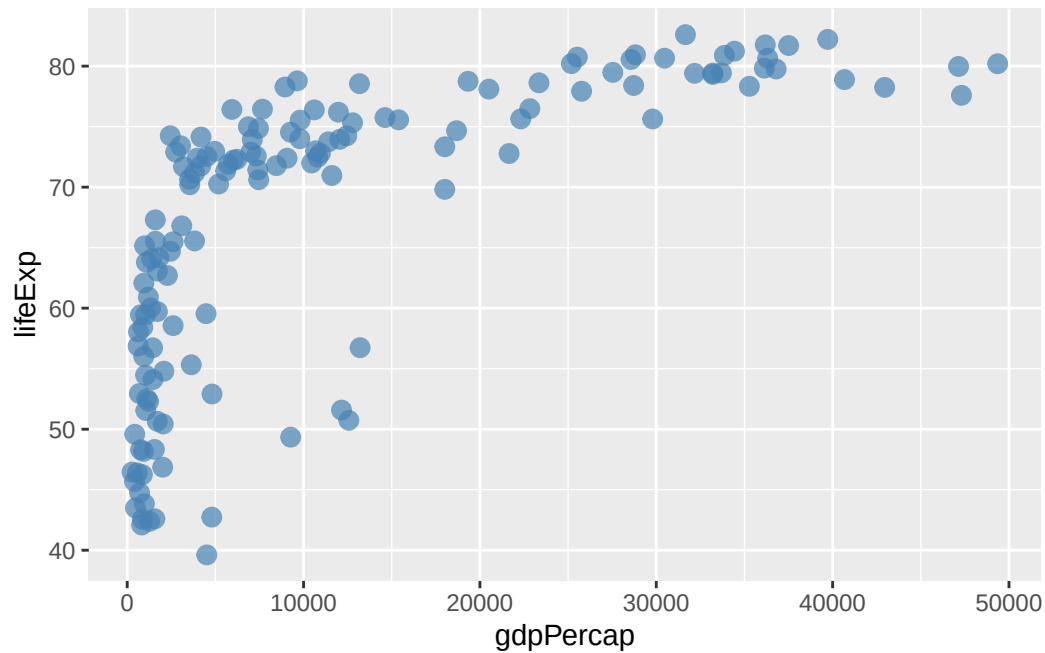
When an aesthetic is mapped, it goes inside `aes()` and represents data. In the next plot, color means continent.

```
p_gap +  
  geom_point(aes(color = continent), size = 3, alpha = 0.7)
```



When an aesthetic is set, it stays fixed and goes outside `aes()`. In the next plot, all points are the same color. The color does not encode information.

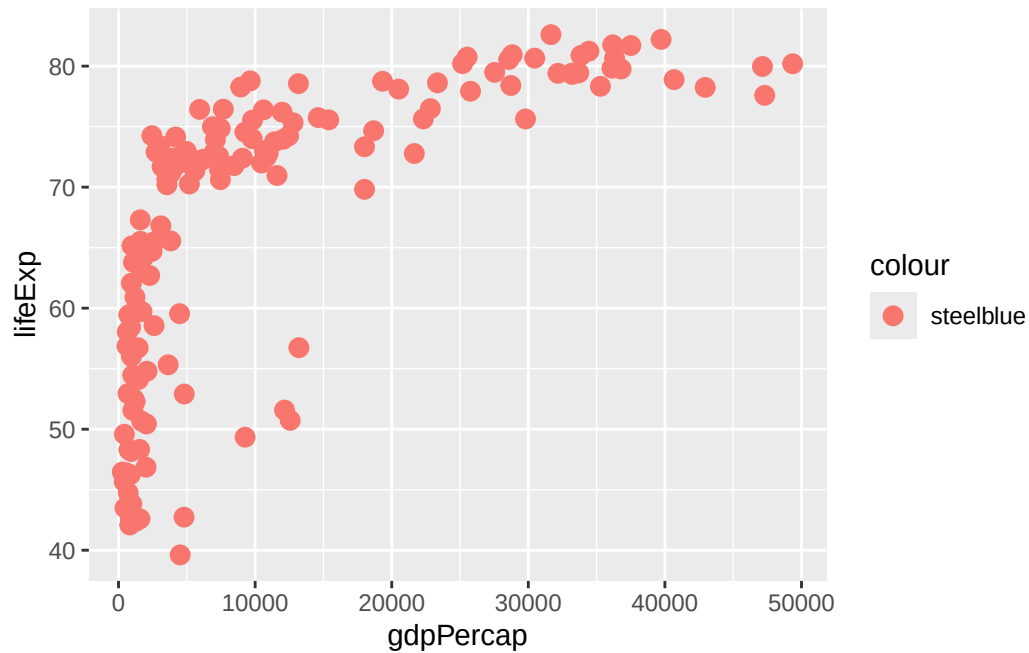
```
p_gap +  
  geom_point(color = "steelblue", size = 3, alpha = 0.7)
```



This distinction matters because viewers interpret visual differences. If color varies because it is mapped to a variable, it carries meaning. If color is fixed, it is a design choice.

A common mistake is to put a fixed color inside `aes()`:

```
p_gap +  
  geom_point(aes(color = "steelblue"), size = 3)
```



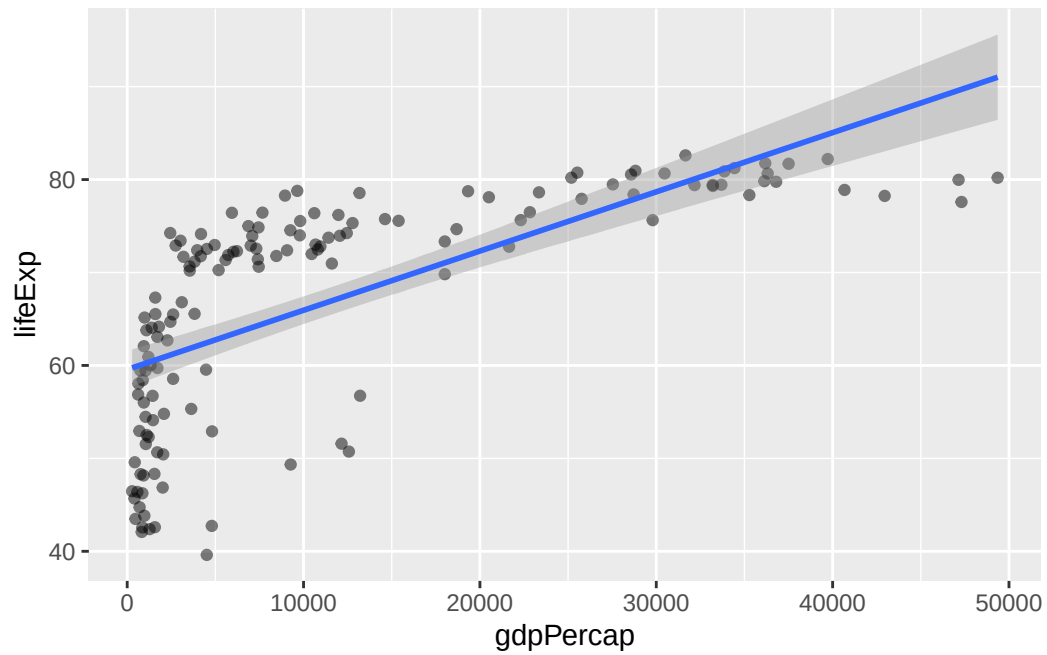
In that plot, "steelblue" is treated like a one-value variable, not as a fixed color instruction. The result is a legend that does not help the reader. The fix is to move fixed settings outside `aes()`.

4.10 Smoothers

Scatterplots show individual observations. Smoothers add a summary of the relationship.

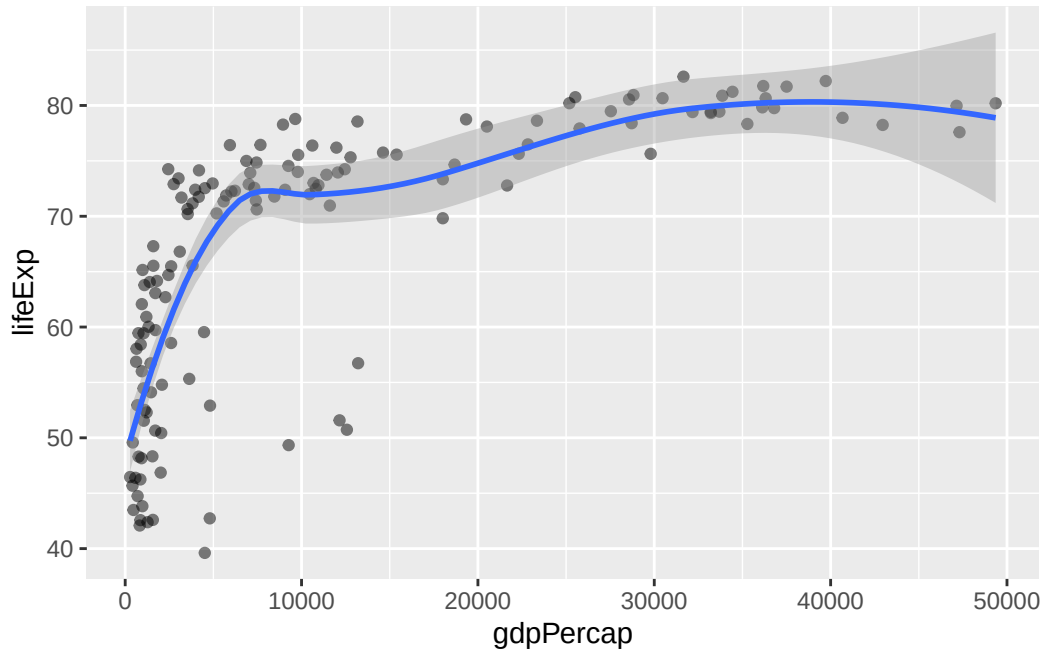
A linear smoother fits a straight line:

```
p_gap +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm", formula = y ~ x)
```



A loess smoother is more flexible:

```
p_gap +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "loess", formula = y ~ x)
```

Neither smoother is automatically correct. A straight line is easy to interpret but may miss curvature. A loess curve can reveal bends in the data but is more descriptive than explanatory. The right choice depends on the purpose of the plot.

4.11 Log Scales

GDP per capita is usually highly skewed. A few very high values can compress most observations into the left side of the plot. A log scale spreads the lower and middle values out while keeping the order intact.

```
p_gap +
  geom_point(alpha = 0.6) +
  scale_x_log10(labels = label_dollar())
```

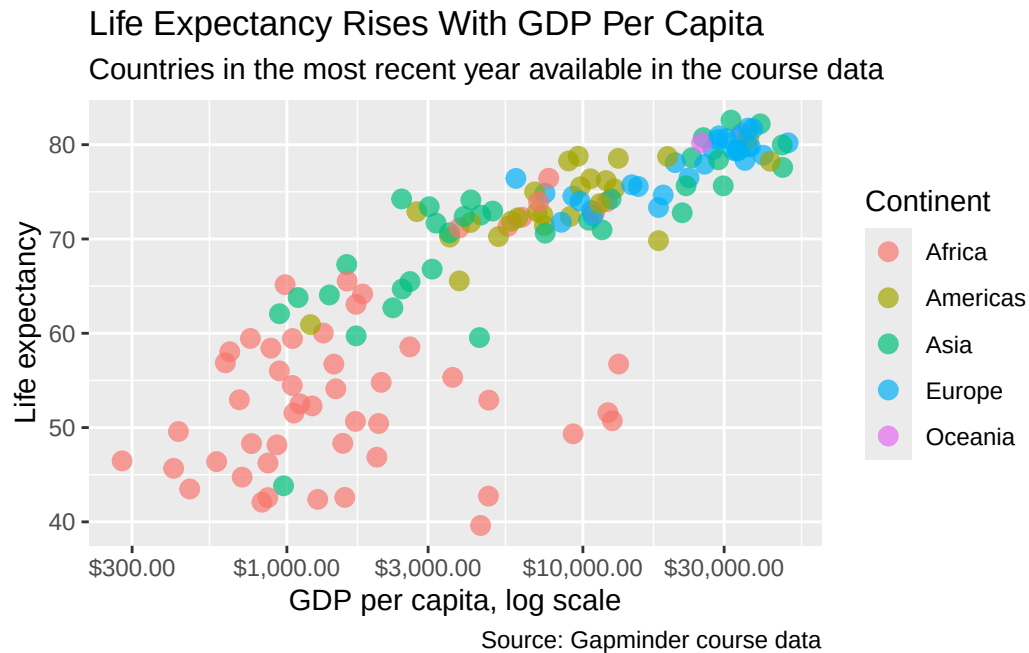


The command `scale_x_log10()` changes the x-axis, not the underlying data object. The argument `labels = label_dollar()` formats the axis as dollars instead of scientific notation.

4.12 Labels With `labs()`

Plots should be understandable outside the code chunk. `labs()` adds titles, subtitles, axis labels, legend labels, and captions.

```
p_gap +
  geom_point(aes(color = continent), alpha = 0.7, size = 3) +
  scale_x_log10(labels = label_dollar()) +
  labs(
    title = "Life Expectancy Rises With GDP Per Capita",
    subtitle = "Countries in the most recent year available in the course data",
    x = "GDP per capita, log scale",
    y = "Life expectancy",
    color = "Continent",
    caption = "Source: Gapminder course data"
  )
```

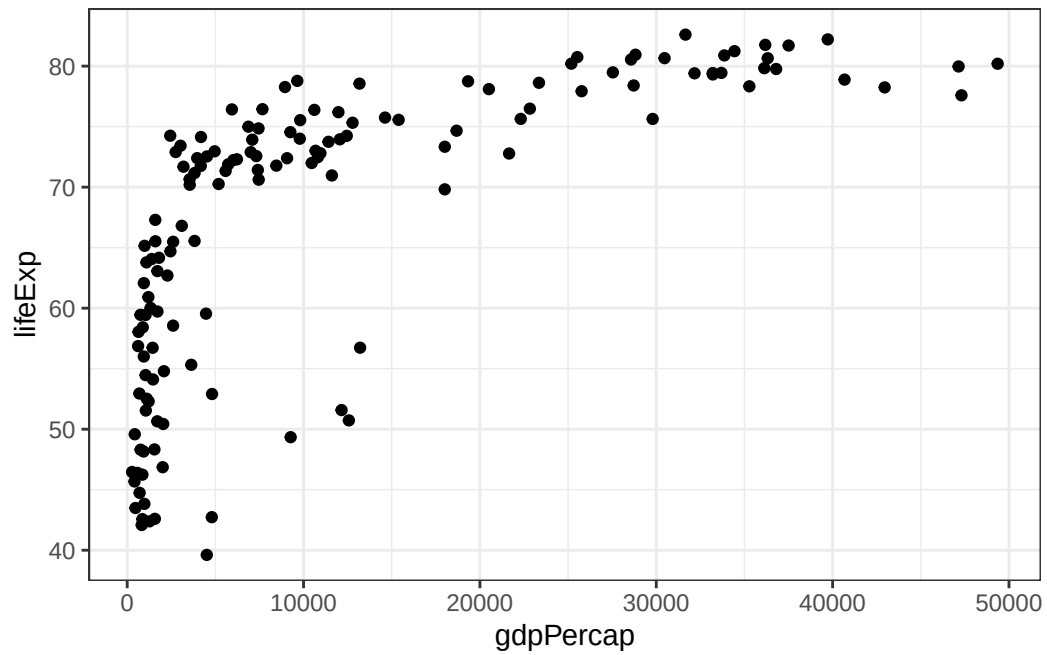


The title tells the reader the main subject. The subtitle adds context. Axis labels state the meaning of the plotted variables. The legend title clarifies the mapped aesthetic. The caption is a natural place for data source information.

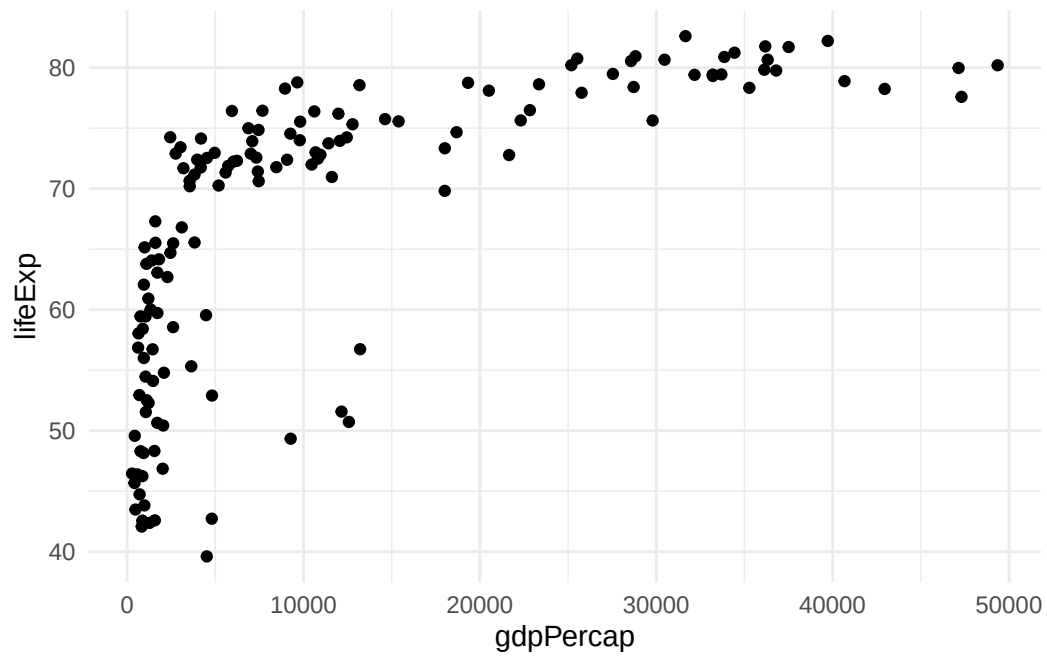
4.13 Themes

Themes control the non-data parts of the plot: background, grid lines, axis text, title placement, legend placement, and similar elements. A theme does not change the data being shown, but it can change how easy the plot is to read.

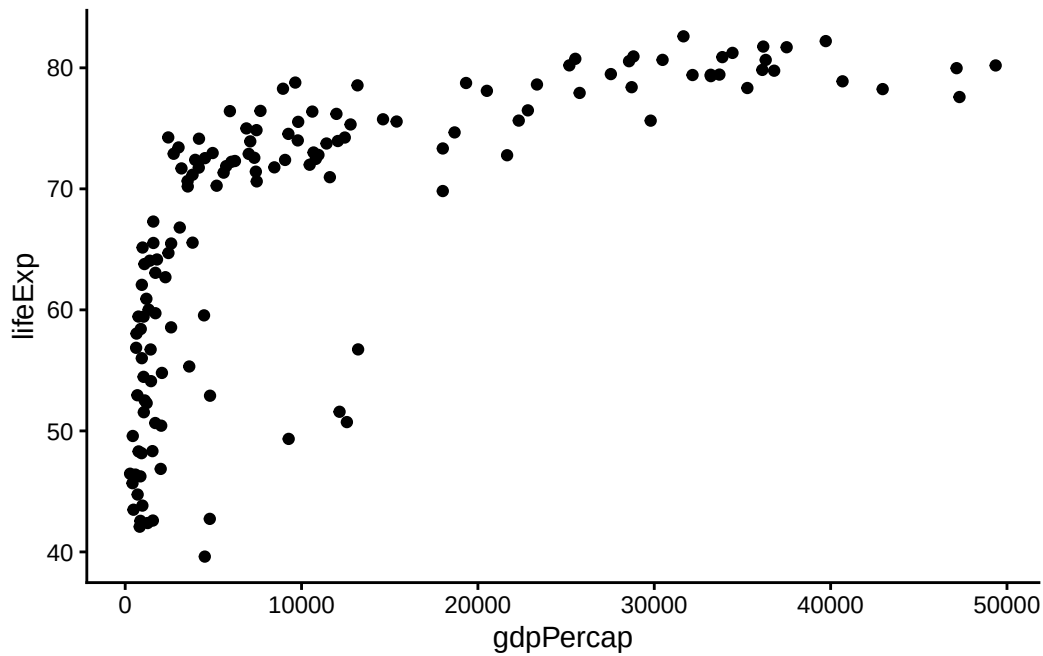
```
p_gap + geom_point() + theme_bw()
```



```
p_gap + geom_point() + theme_minimal()
```

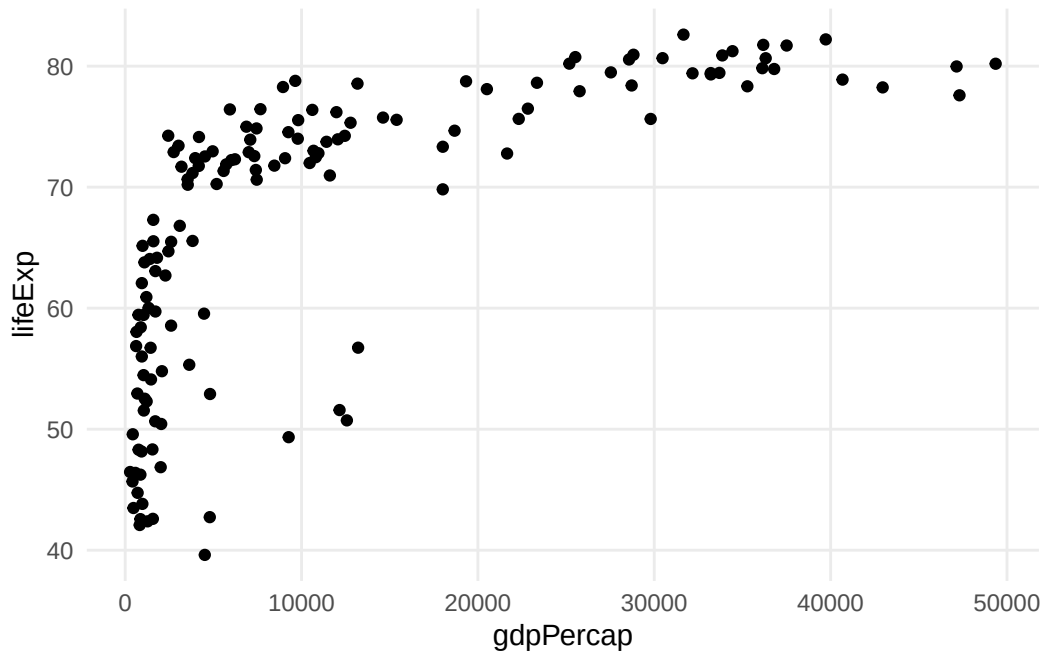


```
p_gap + geom_point() + theme_classic()
```



Built-in themes are good starting points. Individual elements can then be adjusted with `theme()`.

```
p_gap +
  geom_point() +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    plot.subtitle = element_text(size = 11),
    axis.title = element_text(size = 11),
    panel.grid.minor = element_blank()
  )
```



The most important rule is that the theme should support the comparison. A visually dramatic theme that makes the data harder to read is not helping.

4.14 Color Palettes

Default colors are fine for early drafts. For more intentional figures, choose palettes that match the type of data.

- Qualitative palettes distinguish categories with no natural order.
- Sequential palettes show ordered values from low to high.
- Diverging palettes show distance from a meaningful midpoint.

Start from a plot that maps a categorical variable to color:

```
p_gap_color <- p_gap +
  geom_point(aes(color = continent), alpha = 0.75, size = 2) +
  scale_x_log10(labels = label_dollar()) +
  labs(
    x = "GDP per capita, log scale",
    y = "Life expectancy",
    color = "Continent"
  ) +
```

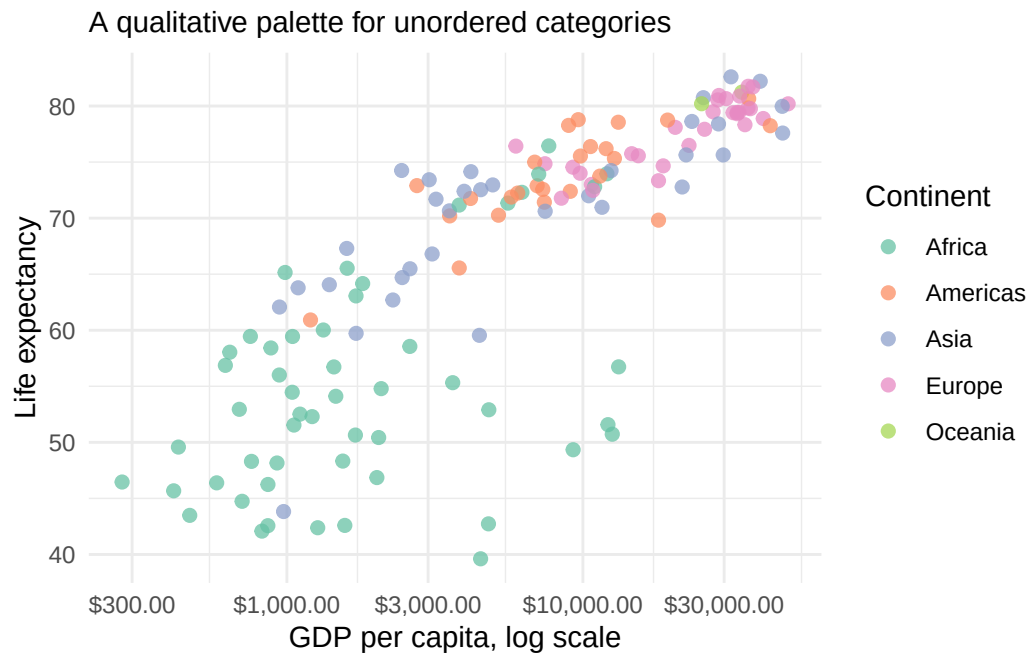
```
theme_minimal()
```

```
p_gap_color
```



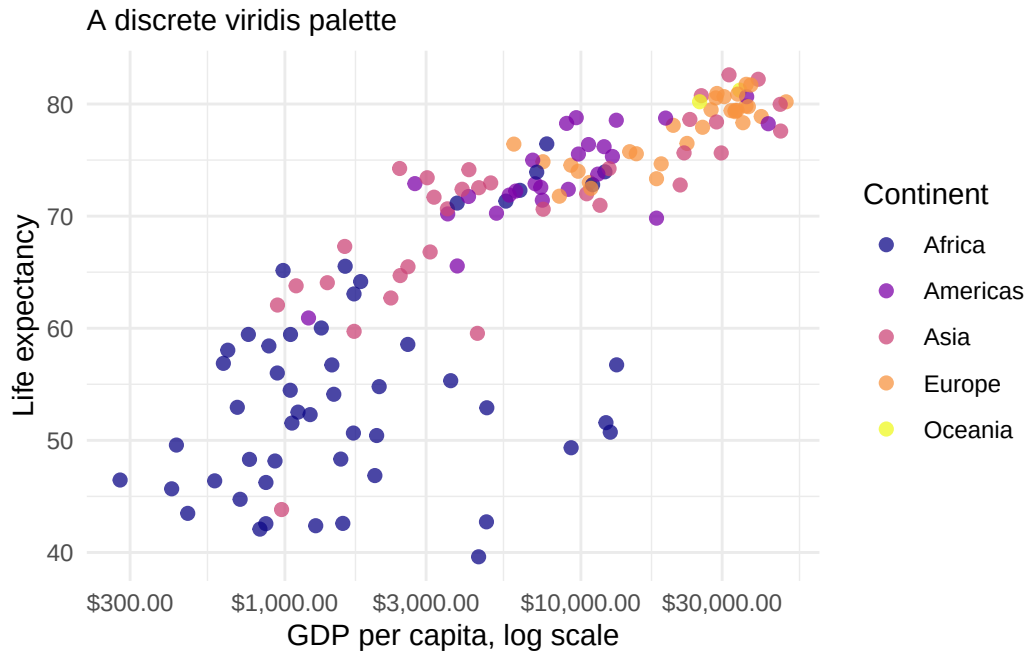
RColorBrewer provides many named palettes.

```
p_gap_color +  
  scale_color_brewer(palette = "Set2") +  
  labs(subtitle = "A qualitative palette for unordered categories")
```



The viridis scales are also useful because they are designed to be readable in color and when converted to grayscale.

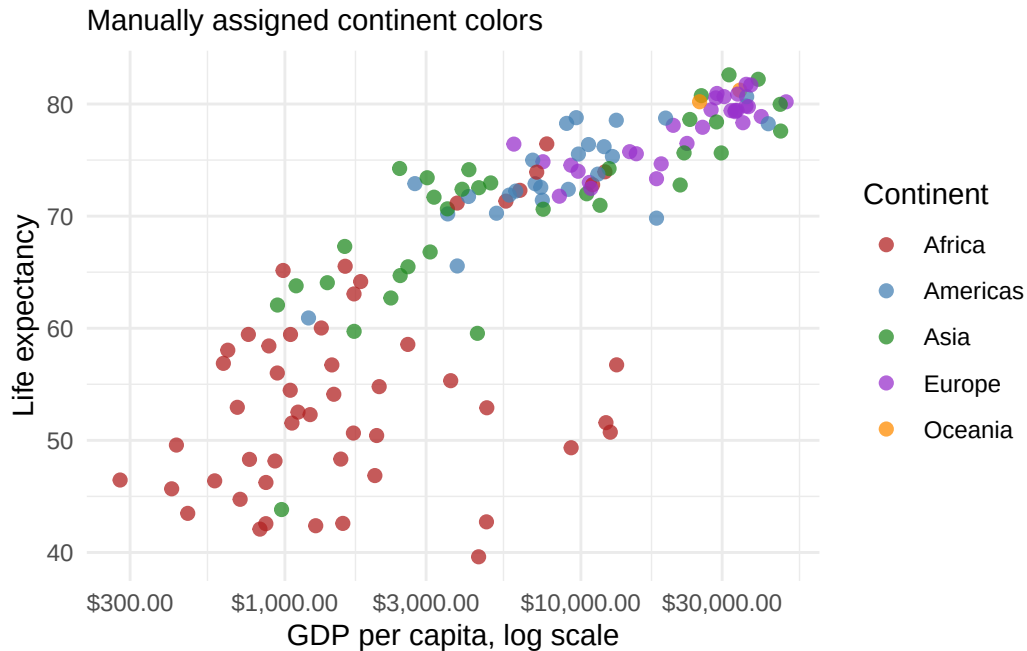
```
p_gap_color +  
  scale_color_viridis_d(option = "plasma") +  
  labs(subtitle = "A discrete viridis palette")
```

Manual color scales are useful when specific categories should keep specific colors across several plots.

```
continent_colors <- c(
  "Africa" = "firebrick",
  "Americas" = "steelblue",
  "Asia" = "forestgreen",
  "Europe" = "darkorchid",
  "Oceania" = "darkorange"
)

p_gap_color +
  scale_color_manual(values = continent_colors) +
  labs(subtitle = "Manually assigned continent colors")
```



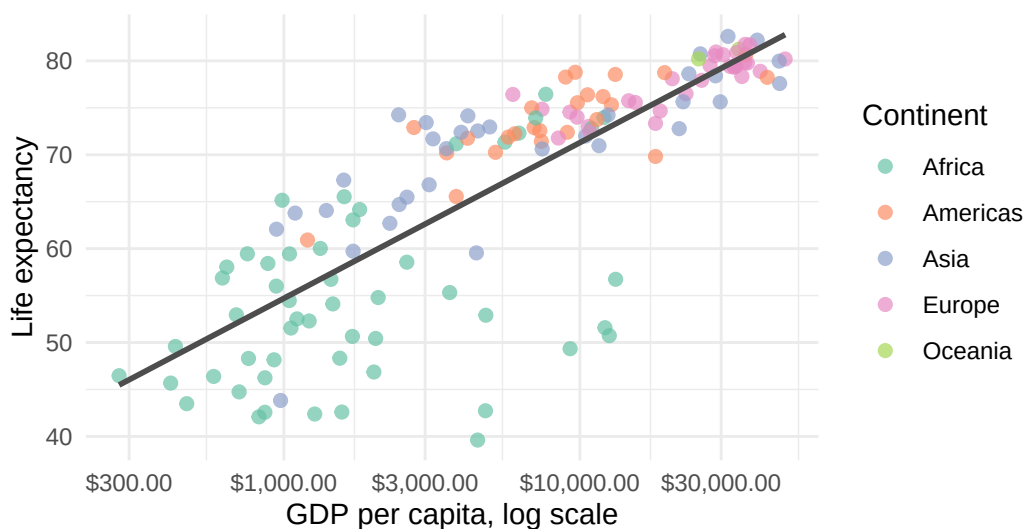
4.15 Putting It All Together

Each piece of this chapter has been introduced as one layer at a time. The full plot stacks them in order: base layer, point geom, smoother, scale, labels, palette, theme.

```
ggplot(
  data = gap_2007,
  mapping = aes(x = gdpPercap, y = lifeExp)
) +
  geom_point(aes(color = continent), alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, color = "gray30") +
  scale_x_log10(labels = label_dollar()) +
  scale_color_brewer(palette = "Set2") +
  labs(
    title = "Life Expectancy Rises With GDP Per Capita",
    subtitle = "Countries in the most recent year available in the course data",
    x = "GDP per capita, log scale",
    y = "Life expectancy",
    color = "Continent",
    caption = "Source: Gapminder course data"
  ) +
  theme_minimal()
```

Life Expectancy Rises With GDP Per Capita

Countries in the most recent year available in the course data



Source: Gapminder course data

Read the code from the top:

- `ggplot(...)` sets the base layer with data and mappings.
- `geom_point(...)` draws each country as a point and maps continent to color.
- `geom_smooth(...)` adds a single linear summary across all continents.
- `scale_x_log10(...)` transforms the x-axis so the lower values are easier to read.
- `scale_color_brewer(...)` swaps the default palette for a qualitative one.
- `labs(...)` adds the title, subtitle, axis labels, legend title, and caption.
- `theme_minimal()` controls the non-data styling.

If the plot needs to change, the line responsible for that change is usually easy to identify.

4.16 Short Exercise

The built-in `mtcars` dataset is always available. Convert it to a tibble first so the car names become a column:

```
cars_tbl <- mtcars |>
  as_tibble(rownames = "car")
```

Using `cars_tbl`, build a scatterplot with:

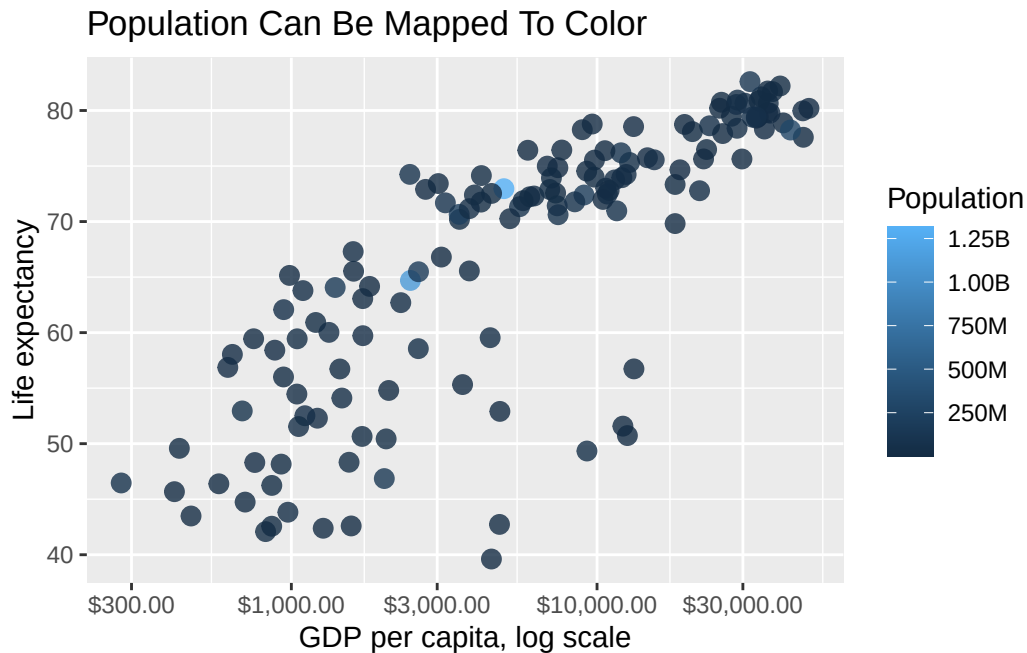
1. `hp` on the x-axis
2. `mpg` on the y-axis
3. `cyl` mapped to color
4. a linear smoother with `se = FALSE`
5. readable labels with `labs()`

```
# Write your code here.
```

4.17 Extras: Continuous Color

Color can represent a continuous variable as well as a category. In the next plot, color represents population.

```
p_gap +  
  geom_point(aes(color = pop), size = 3, alpha = 0.8) +  
  scale_x_log10(labels = label_dollar()) +  
  scale_color_continuous(labels = label_number(scale_cut = cut_short_scale())) +  
  labs(  
    title = "Population Can Be Mapped To Color",  
    x = "GDP per capita, log scale",  
    y = "Life expectancy",  
    color = "Population"  
  )
```

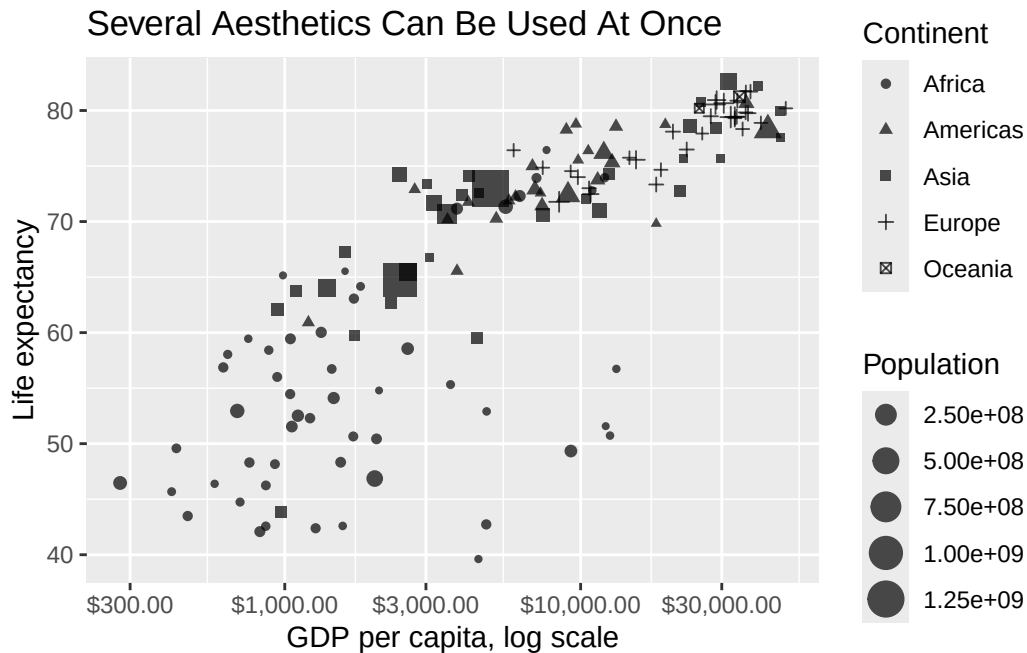


This is possible, but it is not always the best choice. Continuous color scales can be hard to read precisely. Use them when the added variable helps interpretation rather than simply making the plot look more complex.

4.18 Extras: Shape And Size

Other aesthetics can also be mapped or set. Shape works best for a small number of categories. Size can work for quantities, but area comparisons are difficult for readers.

```
ggplot(gap_2007, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(aes(shape = continent, size = pop), alpha = 0.7) +
  scale_x_log10(labels = label_dollar()) +
  labs(
    title = "Several Aesthetics Can Be Used At Once",
    x = "GDP per capita, log scale",
    y = "Life expectancy",
    shape = "Continent",
    size = "Population"
  )
```



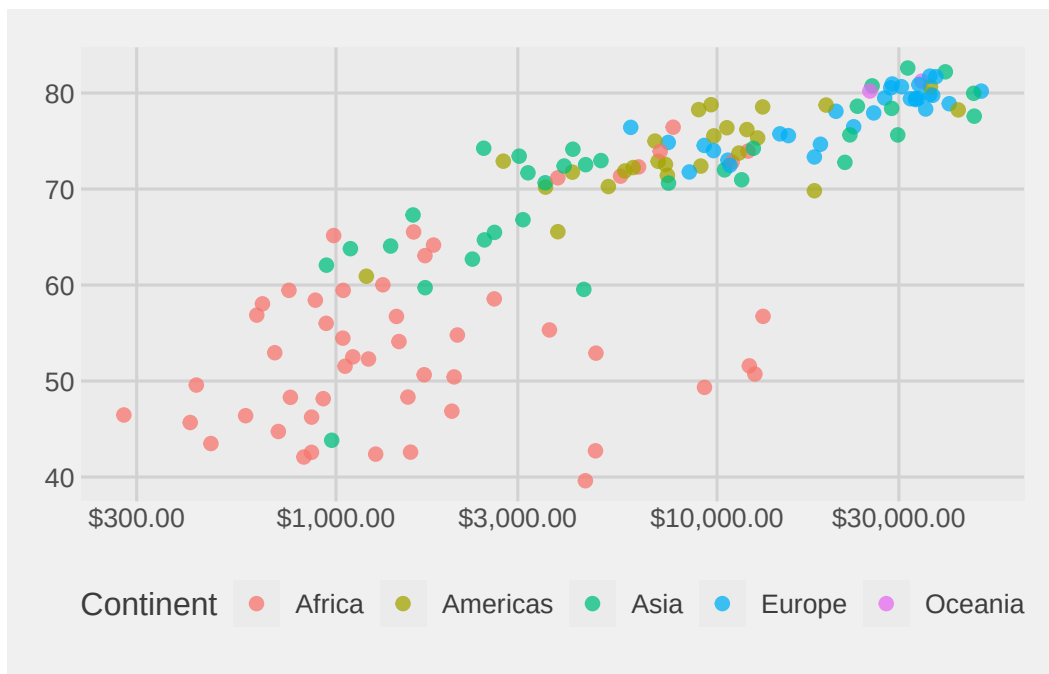
Adding more aesthetics is not automatically an improvement. Each additional encoding asks the reader to do more work. The best plot is usually the one that makes the intended comparison clear with the fewest necessary encodings.

4.19 Extras: External Themes And Rich Text

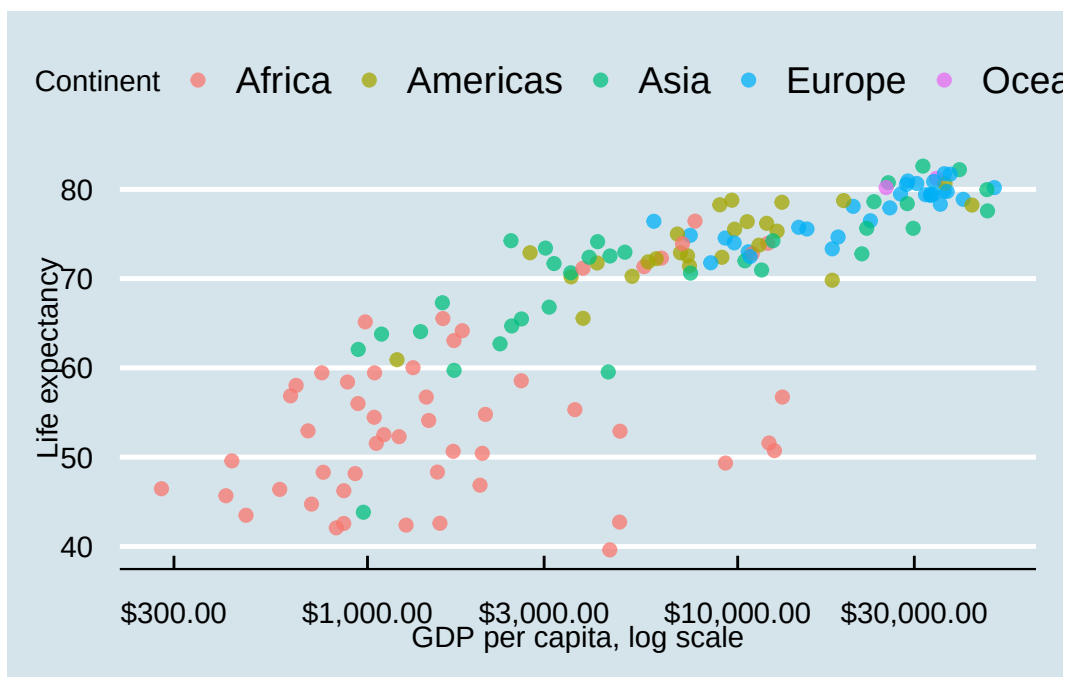
The packages `ggthemes`, `hrbrthemes`, and `ggtext` provide additional styling options. These are useful when a plot needs to match a publication, organization, or report style.

`ggthemes` includes themes that imitate familiar publication and media styles.

```
p_gap_color +
  ggthemes::theme_fivethirtyeight()
```

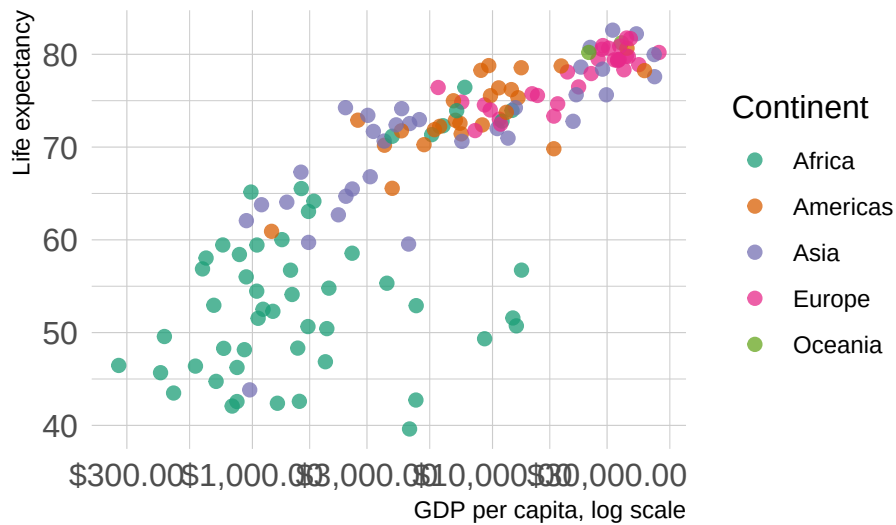


```
p_gap_color +
  ggthemes::theme_economist()
```



The `hrbrthemes` package is especially useful for clean report-style plots.

```
p_gap_color +
  scale_color_brewer(palette = "Dark2") +
  hrbrthemes::theme_ipsum(base_family = "sans")
```

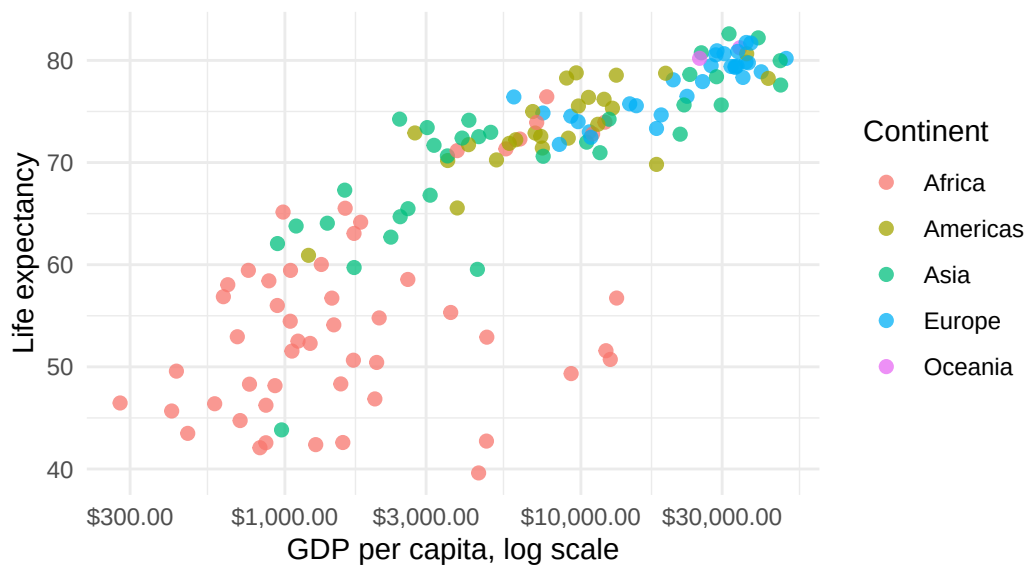


`ggtext` allows limited Markdown and HTML-like formatting inside plot text. This is useful for emphasizing one phrase in a title or coloring part of a subtitle.

```
p_gap_color +
  labs(
    title = "Economic Growth and **Life Expectancy**",
    subtitle = "Countries in <span style='color:darkorange;'>2007</span>"
  ) +
  theme_minimal() +
  theme(
    plot.title = ggtext::element_markdown(),
    plot.subtitle = ggtext::element_markdown()
  )
```


Economic Growth and Life Expectancy

Countries in 2007



Style should refine a figure, not rescue an unclear comparison.

4.20 What Comes Next

The next chapter moves from basic plot construction to visual comparison: separating groups with color, facets, small multiples, free scales, direct labels, and combining plots with `patchwork`. The focus shifts from making a plot work to making a comparison easier to see.